

A powerful new strategy to shape SA's energy future

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BUSINESS DAY 21 August 2012

The National Development Plan, presented to President Zuma last week in parliament, sets out steps to transform South Africa's energy system. By 2030 coal will contribute proportionately less to primary-energy needs, while gas and renewable energy resources – especially wind, solar and imported hydroelectricity – will play a much larger role.

Energy powers economic development and contributes to the welfare of its citizens. But these benefits need to be accomplished within the bounds of environmental sustainability. Approximately 80 per cent of our greenhouse gas emissions are from the energy sector, mainly electricity production from coal.

Some argue that South Africa will not be able to meet its climate change mitigation commitments if it continues to promote electricity-intensive mining and mineral beneficiation industries. However, modelling by the National Planning Commission (NPC) shows that this is indeed possible. Direct, on-site emissions from these sectors are only five per cent of national carbon emissions and will not grow much in the next 20 years

Two-thirds of the carbon emissions associated with mining and minerals processing derive from the electricity they consume. The current integrated resource plan (IRP) envisages that the carbon-intensity of electricity generation will decline by a third by 2030 as renewable energy technologies are introduced. The structure of our minerals sector will also change as gold mining declines, inefficient smelters are upgraded and aluminium plants close at the end of their favourable Eskom contracts.

The National Development Plan shows that our coal, iron, ferro-alloy and platinum sectors could actually grow and total emissions would be lower in 2030 than they are today.

The core challenge in the coal sector is to ensure that the needs of Eskom's existing power stations are met for the remainder of their design life while valuable export revenues are maximised. The majority of South Africa's coal reserves are in the underdeveloped Waterberg field in Limpopo. As coal resources in the central Highveld basin – the location of most of Eskom power stations – diminish, a new heavy haul rail corridor to the Waterberg will need to be developed. The coal line to Richards Bay also needs strengthening to match port export capacity of at least 91 million tons per year by 2020. Expanded coal exports will strengthen our balance of payments and help fund accelerated economic growth.

Coal export markets are changing. We export less to Europe and more to India, which accepts lower ore grades, some of which are needed for Eskom power stations. This has led to calls for restrictions on exports. However, government should be cautious in applying policy measures which might have unintended consequences. For example, banning exports of coal lower than, say, 5500 kcal/kg could dis-incentivise investments in new multi-product mines necessary for supplying future Eskom demand. Instead, a win-win solution between Eskom and the coal miners should be sought. This is entirely possible, as the highest value option for most coal mines is an income stream from both exports *and* Eskom.

South Africa will need to meet about 29 000 megawatts (MW) of new power demand between 2012 and 2030. A further 10 900MW of old Eskom power stations will be retired. The IRP lays out power generation options in a policy-adjusted scenario that seeks a trade-off between least-cost investment, carbon emissions, technology risks, water-use implications, localisation and regional imports. The plan calls for 21 500MW of new renewable energy capacity to be in place by 2030.

International bidding rounds have already been held to fast-track renewable energy procurement with positive outcomes in terms of falling prices and substantial new private investment which will amount to over R100 billion in the next three years. While the prices of solar energy fell by 40 per cent between the first and second bidding rounds, its cost is still more than three times the current average price of electricity generation in South Africa. Further economic modelling is necessary to test the short and medium term sustainability of renewable energy investments.

According to the Department of Energy's IRP, more nuclear energy plants will need to be commissioned from 2023. Although nuclear power provides a low-carbon, base-load alternative to coal, South Africa needs a thorough investigation of the implications of nuclear energy, including its costs, financing options, institutional arrangements, safety, environmental costs and benefits, localisation and employment opportunities, and uranium enrichment and fuel-fabrication possibilities.

While many of these issues were investigated in the IRP, and in a recent International Atomic Energy Agency sanctioned Integrated Nuclear Infrastructure Review, a nuclear fleet of power stations will involve a level of investment unprecedented in South Africa. An in-depth investigation into the financial viability of nuclear energy is thus vital. The National Nuclear Energy Executive Coordinating Committee, chaired by the Deputy-President, will have to make a final "stop-go" decision on South Africa's nuclear future after actual costs and financing options are revealed.

South Africa needs a "plan B" should nuclear energy prove too expensive, sufficient financing be unavailable or timelines too tight. All possible alternatives need to be explored, including regional hydropower and greater use of gas which could provide reliable and flexible power generation through combined-cycle gas turbines. Gas power plants have lower scale-economies and smaller plants can be built quickly to match actual demand growth. While the operational costs of gas power are arguably higher than those of nuclear stations, their unit capital costs are cheaper, they are more easily financed and they are more able to adjust their output to make up the shortfall from variable renewable energy sources.

Gas supplies from Karoo shale, even if economically viable and environmentally acceptable, will take many years to develop. In the meantime, West Coast gas resources should be developed and investments should be made in liquefied natural gas (LNG) import infrastructure as insurance for the future.

Further refinements and regular updates of the IRP will be necessary to track electricity demand (which could be lower because of energy-efficiency gains, sharp price increases and a sluggish economy, or higher if economic growth accelerates) and to assess whether new generation technologies are delivering timely and affordable power. Planning capability needs to be vested in an independent system and market operator (ISMO) which should be spun out of Eskom. The ISMO will also be responsible for procurement and contracting of new power including independent power producers (IPPs). To ensure a level playing field between Eskom and IPPs, transmission assets and operations should be transferred to the ISMO. Remaining regulatory uncertainties that need to be resolved include the question of IPPs selling to customers other than Eskom, fair access to the electricity grid and rights to trade electricity.

The National Development Plan addresses also other energy policy and planning issues including energy demand management, electricity prices, electrification targets, reliability and maintenance backlogs in municipal electricity distribution networks, the desirability and timing of petroleum refinery investments and modal shifts in transport.

Planning for our energy future is vital for the prosperity and wellbeing of South Africans. It involves complex linkages and alignment with economic, social and environmental policies, strategies, programmes and investments. The National Planning Commission stands ready to support the Department of Energy, other government departments, industry and civil society to ensure that our energy sector delivers reliable, competitively priced and environmentally sustainable services to our economy and society.

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